4.5 DeMoivre's Theorem

Pre-Calculus 4.5 · Student sheet

Name _____ Period ____ Date _____

I can…

- use DeMoivre's Theorem to raise a complex number to a power.
- find the nth roots of a complex number.
- solve equations like $x^3 = 1$ using roots of complex numbers.

Vocabulary · fill in the missing word

DeMoivre's Theorem — $[r(\cos\theta + i\sin\theta)]^n =$ _____

nth root of a complex number — a number u with $u^n =$ _____

Part 1 · Powers of 1 + i

Multiply the long way. Then compare with $1 + i = \sqrt{2}(\cos\pi/4 + i\sin\pi/4)$.

Power	The long way	$(\sqrt{2})^n$	$n \cdot \pi/4$
$(1 + i)^2$			
$(1 + i)^4$			

Part 2 · Notes and examples (fill in each note, then work the example)

DeMoivre's Theorem: $[r(\cos\theta + i\sin\theta)]^n =$ _____$(\cos n\theta + i\sin n\theta)$

Example 1

Find $(-1 + \sqrt{3}\,i)^{12}$.

Example 2

Find $(1 + i)^8$.

nth roots: $z = r(\cos\theta + i\sin\theta)$ has exactly n distinct nth roots: $\sqrt[n]{r}[\cos((\theta + 2\pi k)/n) + i\sin((\theta + 2\pi k)/n)]$, $k = 0, 1, \ldots,$ _____.

Example 3

Find the cube roots of 8.

Spacing: the nth roots lie on a circle, equally spaced _____ apart.

Example 4

Find the cube roots of $-2 + 2i$. Round to four decimal places.

Example 5

Solve $x^3 - 1 = 0$.

Part 3 · Partner practice

Partner 1

Find $(\sqrt{3} + i)^4$ in standard form.

Partner 2

Find the square roots of $4i$.